IMPLEMENTATION PLAN v2 · BUILD-READY · 15 OCT 2026

KJS India Platform Implementation Plan v2

Next.js (SEO) + FastAPI + PostgreSQL + Claude (Anthropic API)
UI designs, backend logic, file structure, nightly jobs, sprint plan

Builder: Vineet Pandey with Claude Code · Supersedes wireframe_kjs2.pdf and research_01_implementation_plan.pdf
Companions: ui_designs_v3.html (15 new screens), kjsKoko.html (portal screens), research_01.pdf (sources + feasibility)
Platform go-live: Fri 11 Dec 2026 · Buyer Intelligence: Jan - Jun 2027 (gated phases)
Rule: build one module at a time, end to end (DB, API, UI, tests), then commit.

Contents

1 Scope and decisions
2 Architecture
3 Tech stack
4 Repository file structure
5 Frontend: Next.js routes & rendering
6 SEO implementation
7 Design system & UI designs
8 Backend: module layout & conventions
9 Data model
10 API endpoints
11 Business logic (algorithms)
12 Jobs & nightly scheduler
13 Claude (Anthropic API) design
14 Source connectors & feasibility
15 Security, privacy, compliance
16 DevOps, environments, monitoring
17 Testing strategy
18 Delivery plan with dates
19 Claude Code task prompts
20 Open items

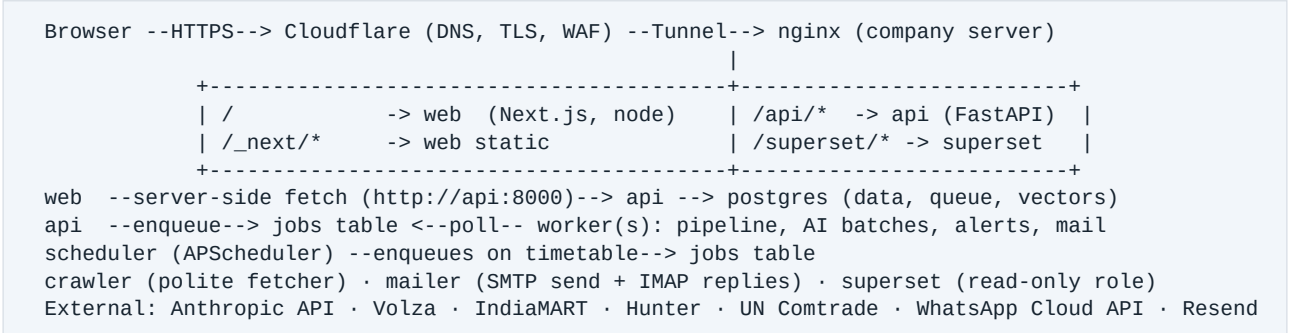
1 Scope and decisions

One platform, three faces, built in two stages. **Stage 1 (now - 11 Dec 2026):** public website, Koko AI, employee portal (attendance, tasks, auto-assign, SLA leads), admin console (order boards, alerts, approvals, audit, insights). **Stage 2 (Jan - Jun 2027):** Buyer Intelligence (sources, nightly pipeline, scoring) and Outreach (approved email sequences), then samples, quotes and partners.

Decision	Choice	Reason
Frontend	**Next.js (App Router, TypeScript)** for public site, portal and admin in one app	Server rendering + static generation for SEO; one codebase, one design system
Backend	FastAPI (Python 3.12), SQLAlchemy 2, Alembic, Pydantic v2	Python is best for data pipelines, pandas, crawling and the Anthropic SDK
Database	PostgreSQL 16 + pg_trgm + pgvector	Fuzzy de-dupe, Koko embeddings, job queue, all in one
Jobs	Postgres queue (SKIP LOCKED) + APScheduler	No Redis/Celery to run
AI	Anthropic API, default model claude-opus-5-5; Batch API at night	Quality first; 50% batch price; cheaper model only after a test (section 13)
Analytics	Apache Superset embedded	Charts without hand-coding
Hosting	Docker Compose on the company server + Cloudflare Tunnel	No open ports; free HTTPS; easy move to cloud
Category gate	Catalogue open; gate price sheet / spec / sample / quote	Research_01 recommendation; switchable per category in admin if management keeps the form
Languages	Public: EN, AR, ES, FR (RTL for AR); portal/admin: EN	Target markets; Arabic RTL from day one in layout

Change from plan v1.1: React + Vite is replaced by Next.js. Effect on the 11 Dec date: about +3 working days in Week 1 (Next.js setup, i18n routing, SEO plumbing), recovered by building portal screens with the same components. If Week 2 slips, Superset moves to the buffer week.

2 Architecture



Concern	How it works
Same-origin cookies	Browser talks only to www.kjsindia.com; nginx routes /api to FastAPI, so the httpOnly session cookie works for portal/admin without CORS
Server rendering	Next.js server components call FastAPI on the internal network (http://api:8000) with a service token for public content
Auth	FastAPI owns login, sessions, roles (admin / manager / employee / partner / customer). Next.js middleware redirects to login if the session cookie is missing; layouts call /api/auth/me for role
Content updates	When admin edits a product, page or offer, FastAPI calls Next.js /api/revalidate with a secret and a tag, so cached pages refresh within seconds (ISR on demand)
Files	Uploads stored on a /srv volume via FastAPI; product images served through next/image from /media (cached by Cloudflare)

3 Tech stack

Layer	Choice	Notes
Web framework	Next.js, App Router, TypeScript (strict)	Pin the current stable version at setup (create-next-app@latest); output: standalone for Docker
UI	Tailwind CSS + shadcn/ui (Radix) + lucide-react icons	Design tokens from section 7; RTL via logical classes (ms-/me-)
i18n	next-intl (locale routing, messages in /messages/*.json)	localePrefix always: /en, /ar, /es, /fr
Data (portal/admin)	TanStack Query + typed client generated from FastAPI OpenAPI (openapi-typescript + openapi-fetch)	No hand-written API types
Forms	react-hook-form + zod	Same validation rules mirrored in Pydantic
Tables & charts	TanStack Table; Superset embed for analytics; Recharts for small inline charts	
API	FastAPI, SQLAlchemy 2 (async), Alembic, Pydantic v2, structlog	
Data libs	pandas, openpyxl, phonenumbers, pycountry, email-validator, httpx, BeautifulSoup, Playwright (allow-list only)	
AI	anthropic (official Python SDK)	Messages, Message Batches, structured outputs, web search tool
Embeddings (Koko)	Open-source multilingual model (e.g. multilingual-e5-small, 384-d) in the worker + pgvector	
Email	Resend/Brevo for system mail; SMTP/IMAP mailboxes for outreach	Separate sending domain for outreach
WhatsApp	WhatsApp Business Platform (Cloud API) - templates for alerts	Opt-in staff only
Testing	pytest, Vitest, Playwright E2E, Lighthouse CI	
Ops	Docker Compose, GitHub Actions, Sentry, UptimeRobot, restic backups to B2/R2	

4 Repository file structure

One monorepo. Claude Code reads CLAUDE.md at the root plus a short CLAUDE.md in each app.

```
kjs-platform/
|- CLAUDE.md                    project memory: stack, commands, rules
|- Makefile                     make up | down | test | lint | migrate | seed | gen-api
|- docker-compose.yml           dev stack;  deploy/compose.prod.yml for server
|- .env.example                 every setting with a fake value
|- docs/  PRD.md  ARCHITECTURE.md  PROGRESS.md  BACKLOG.md  research/ (research_01, ui designs)
|- apps/
|  `- web/                      Next.js app
|     |- app/
|     |  |- [locale]/
|     |  |  |- (public)/
|     |  |  |  |- page.tsx                          P1 home
|     |  |  |  |- products/[category]/page.tsx      P2
|     |  |  |  |- products/[category]/[slug]/page.tsx  P3
|     |  |  |  |- private-label/page.tsx            configurator full page
|     |  |  |  |- importers/[country]/[product]/page.tsx  P4
|     |  |  |  |- partners/page.tsx                 P6
|     |  |  |  |- about/  contact/  privacy/  layout.tsx
|     |  |  |  `- layout.tsx                        html lang/dir, fonts, header, footer
|     |  |  |- portal/                              employee app (EN only)
|     |  |  |  |- page.tsx (E1 My day)  attendance/  tasks/  buyers/[id]/ (E2)
|     |  |  |  |- order-boards/[code]/  queries/  leads/  layout.tsx (auth guard, sidebar)
|     |  |  |- admin/
|     |  |  |  |- page.tsx  insights/  team/  approvals/  auto-assign/  order-boards/
|     |  |  |  |- bi/ sources/ (A1)  markets/ (A2)  discovered/ (A3)  review/ (A4)  buyers/
|     |  |  |  |- outreach/ approvals/ (A5)  campaigns/ (A6)  mailboxes/
|     |  |  |  |- website/ offers/  products/  pages-seo/  koko/   settings/  audit/
|     |  |  |- employee/login/page.tsx
|     |  |  |- api/revalidate/route.ts              on-demand ISR (secret)
|     |  |  |- sitemap.ts   robots.ts   not-found.tsx   manifest.ts
|     |  |- components/ ui/ (shadcn)  public/  portal/  admin/  koko/  seo/ (JsonLd.tsx)
|     |  |- lib/ api/ (generated client + fetchers)  auth.ts  seo.ts  format.ts  i18n.ts
|     |  |- messages/ en.json  ar.json  es.json  fr.json
|     |  |- middleware.ts            locale routing + auth redirect for /portal, /admin
|     |  |- next.config.ts  tailwind.config.ts  Dockerfile
|- services/
|  `- api/                      FastAPI
|     |- app/
|     |  |- main.py   core/ (config, db, security, sessions, logging, errors, deps)
|     |  |- modules/<name>/ router.py schemas.py models.py service.py tests/
|     |  |    auth users catalogue content leads koko attendance tasks assign sla
|     |  |    orderboards files alerts approvals audit analytics partners samples quotes
|     |  |    bi_sources bi_markets bi_buyers bi_review bi_discovery outreach
|     |  |- jobs/ queue.py worker.py scheduler.py registry.py  tasks/*.py (one per job)
|     |  |- ai/ client.py batch.py discovery.py prompts/*.md schemas/*.py evals/
|     |  |- connectors/ base.py comtrade.py volza.py indiamart.py rfq_mailbox.py
|     |  |              hunter.py file_inbox.py places.py
|     |  |- crawler/ robots.py frontier.py fetch.py extract.py
|     |  |- mailer/ send.py imap.py warmup.py policy.py templates/
|     |  `- normalise/ country.py phone.py email.py company.py
|     |- migrations/ (alembic)   tests/ (integration)   pyproject.toml   Dockerfile
|- deploy/ nginx.conf  cloudflared.yml  backup.sh  restore.sh  deploy.sh  superset/
`- .github/workflows/ ci.yml (lint, types, tests, build)  deploy.yml (tag -> server)
```

5 Frontend: Next.js routes and rendering

Route	Screen	Rendering	Data	Cache / revalidate
/[locale]	P1 Home	Server component, static	offers, best sellers, categories	ISR, tag "home"
/[locale]/products/[category]	P2	Static per category x locale (generateStaticParams)	category + products	tag "cat:{slug}"
/[locale]/products/[category]/[slug]	P3	Static; configurator + gate modal are client components	product, specs, price bands	tag "product:{slug}"
/[locale]/importers/[country]/[product]	P4	Static only for published pages; others 404	landing page copy + market facts	tag "landing:{id}"
/[locale]/partners	P6	Static	content	tag "content"
/employee/login	Login	Client	POST /api/auth/login	no cache
/portal/*	E1, E2 + kjsKoko portal screens	Client components behind auth	TanStack Query to /api	no cache
/admin/*	A1-A7 + kjsKoko admin screens	Client components behind auth + role	TanStack Query to /api	no cache
/sitemap.xml, /robots.txt	SEO	Route handlers	published URLs from API	regenerated daily + on publish

Key frontend rules

- Public pages never block on the browser: no client-side data fetching for content; Koko loads lazily after the page (dynamic import, no SSR).
- Locale layout sets `<html lang dir>`; Arabic uses dir="rtl" and Tailwind logical classes so one stylesheet serves both.
- All text through next-intl messages; product content comes translated from the API (reviewed translations only; else English with an EN badge).
- Forms: react-hook-form + zod; submit to /api with the CSRF token header; show server validation errors inline.
- Portal/admin share the sidebar component; menu items come from one config filtered by role (fixes the uneven sidebars from earlier drafts).

6 SEO implementation

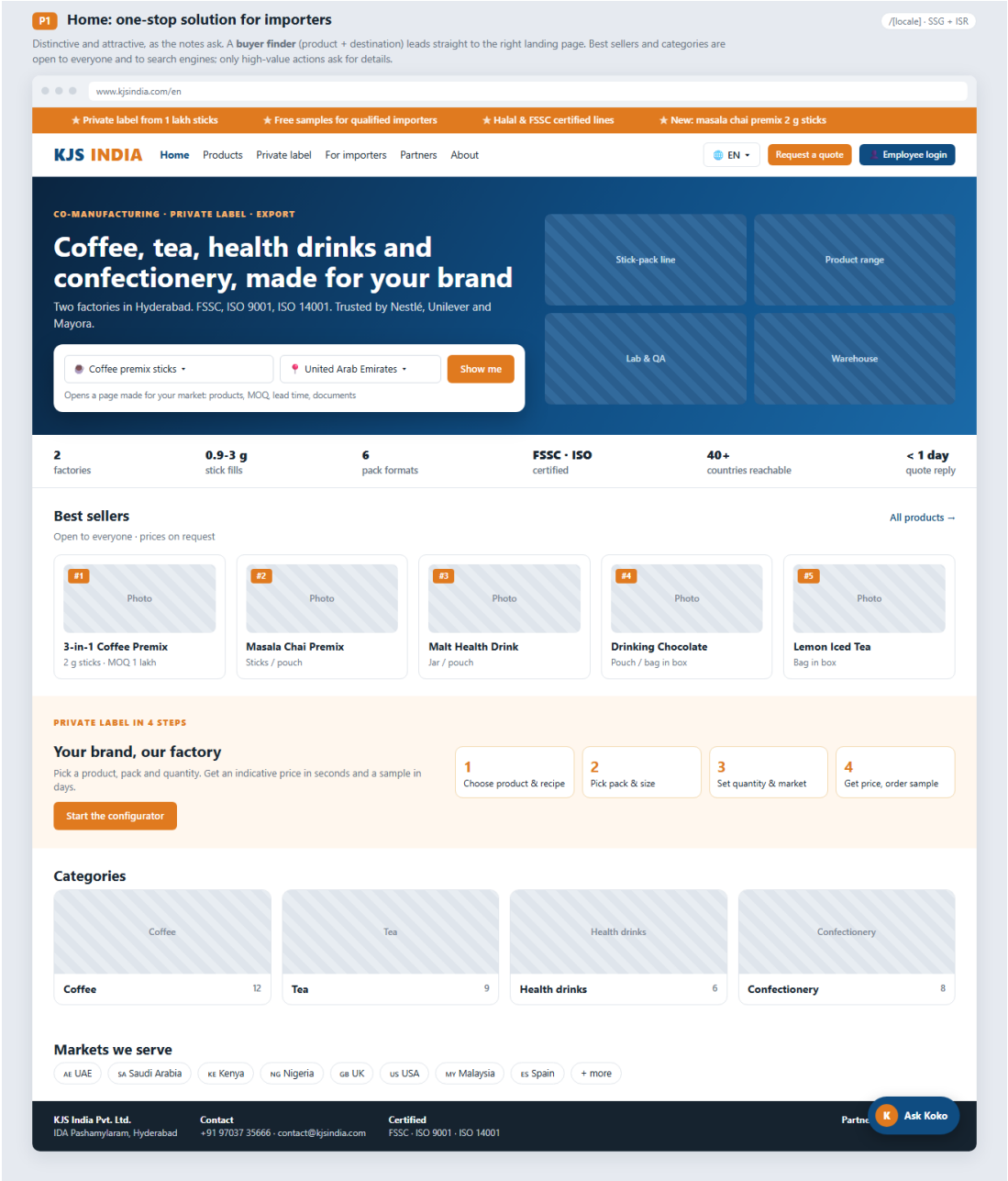
Item	Implementation
Metadata	generateMetadata on every public route: title (<= 60 chars), description (<= 155), canonical, alternates.languages for hreflang (en, ar, es, fr, x-default)
Sitemap	app/sitemap.ts reads /api/public/sitemap: home, categories, products, published landing pages, per locale, with lastModified
Robots	app/robots.ts: allow public; disallow /portal, /admin, /api, /employee; link to sitemap
Structured data	JsonLd component: Organization (home), BreadcrumbList (all), Product without price (P3), FAQPage (P4), WebSite with SearchAction
Social cards	Open Graph + Twitter tags; generated OG images per product with next/og
Performance	next/image (WebP/AVIF, sizes), next/font (self-hosted Inter + Noto Sans Arabic), no layout shift; target LCP < 2.5 s, CLS < 0.1, INP < 200 ms; Lighthouse CI >= 90 on P1-P4
Programmatic pages (P4)	Created as drafts by the system; noindex until a person reviews copy and translation; each needs unique facts (Comtrade numbers, documents, FAQs); max ~50 at launch
Gate and SEO	Gate covers actions (price sheet, spec, sample, quote), never page content, so everything stays indexable

Item	Implementation
Measurement	Google Search Console + Bing Webmaster; privacy-friendly analytics (self-hosted Umami/Plausible) with consent where required

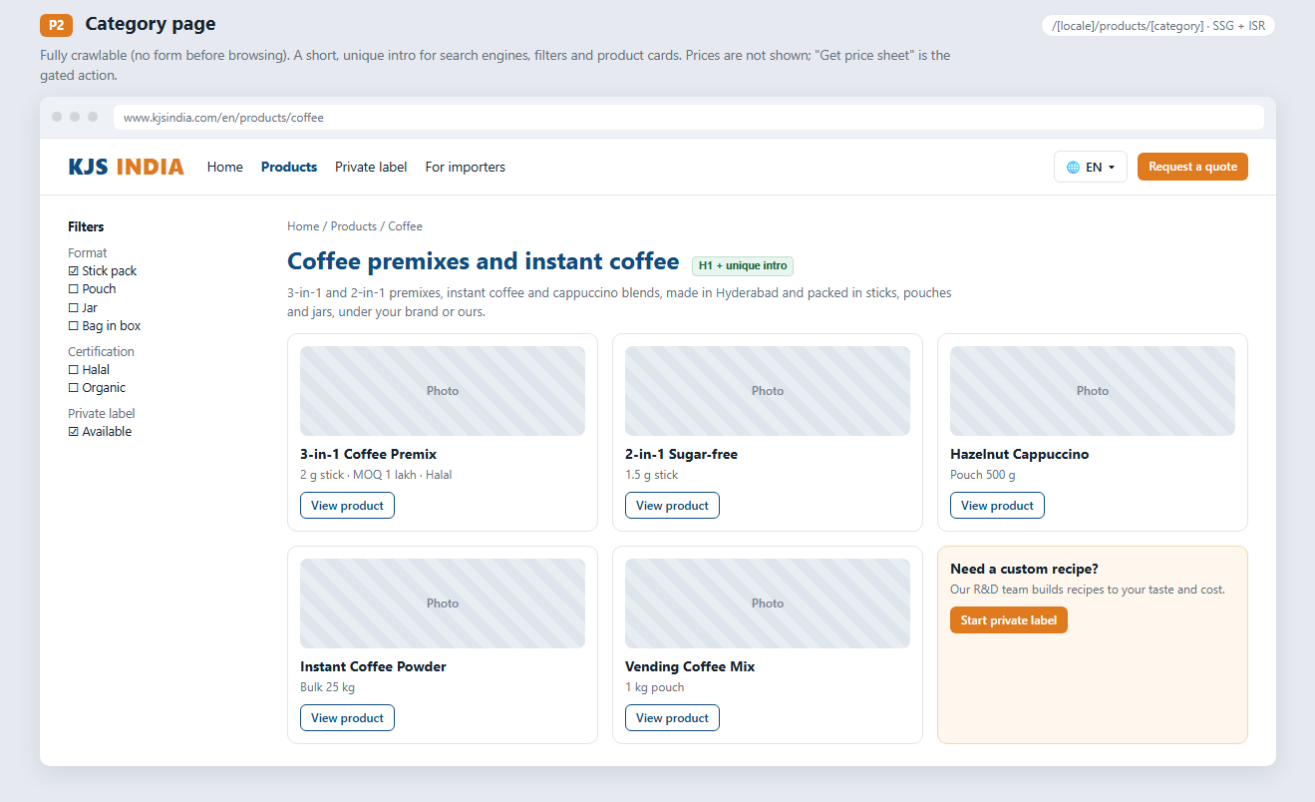
7 Design system and UI designs

Token	Value	Use
Brand	#0F4C81	Header, primary links, admin accents
Accent	#E07A1F	Primary CTA (quote, sample), ticker, step numbers
Ink / muted	#16232F / #66727F	Text / secondary text
Surface / line	#F4F6F9 / #E1E6EC	Page background / borders
Status	green #1F8A4C, amber #D99A00, red #C0392B	SLA, verdicts, alerts (always with a label)
Type	Inter (UI), Noto Sans Arabic (AR); 14-16 px body, 24-36 px headings	
Shape	Radius 8-12 px; cards with 1 px line, no heavy shadows	
Components	shadcn/ui: Button, Input, Select, Dialog, Sheet, Tabs, Table, Badge, Toast, Command (search)	

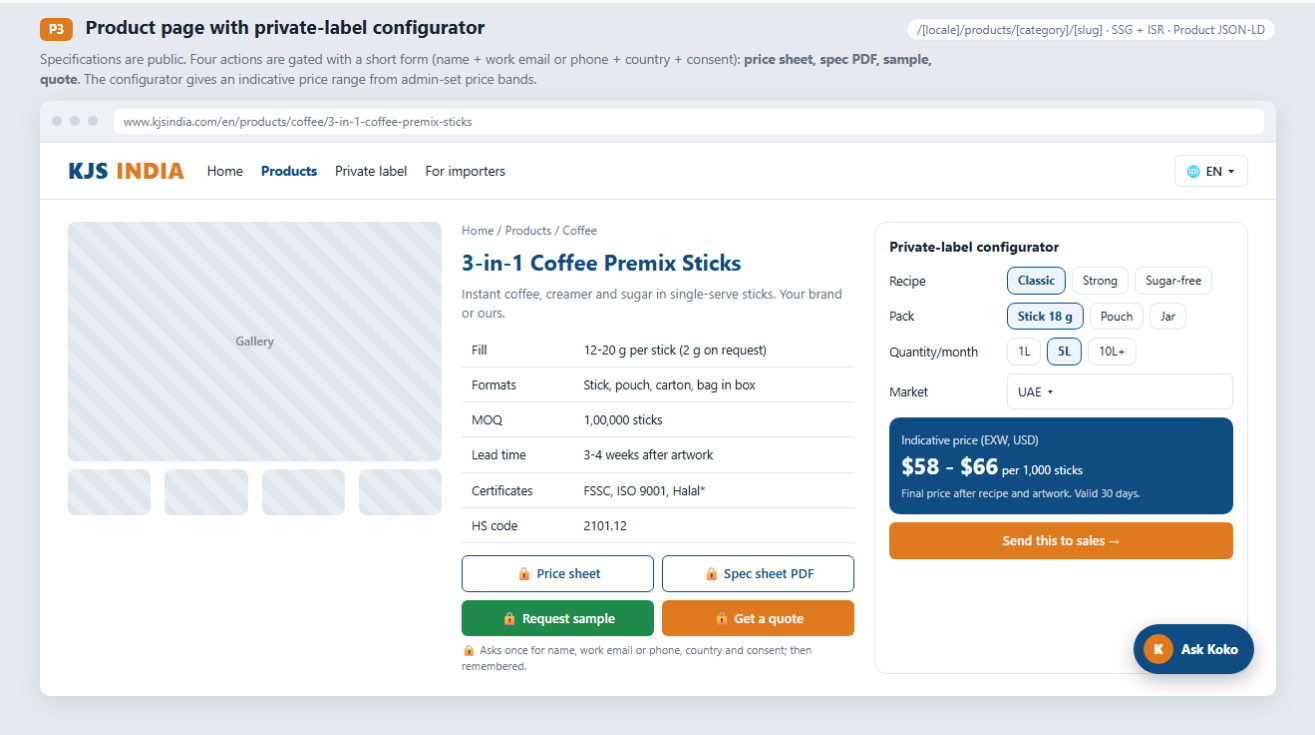
The 15 new screens below are in **ui_designs_v3.html** (open in a browser; hover the charts). Portal and admin screens for attendance, tasks, auto-assign, order boards, approvals and team are unchanged from **kjsKoko.html**.



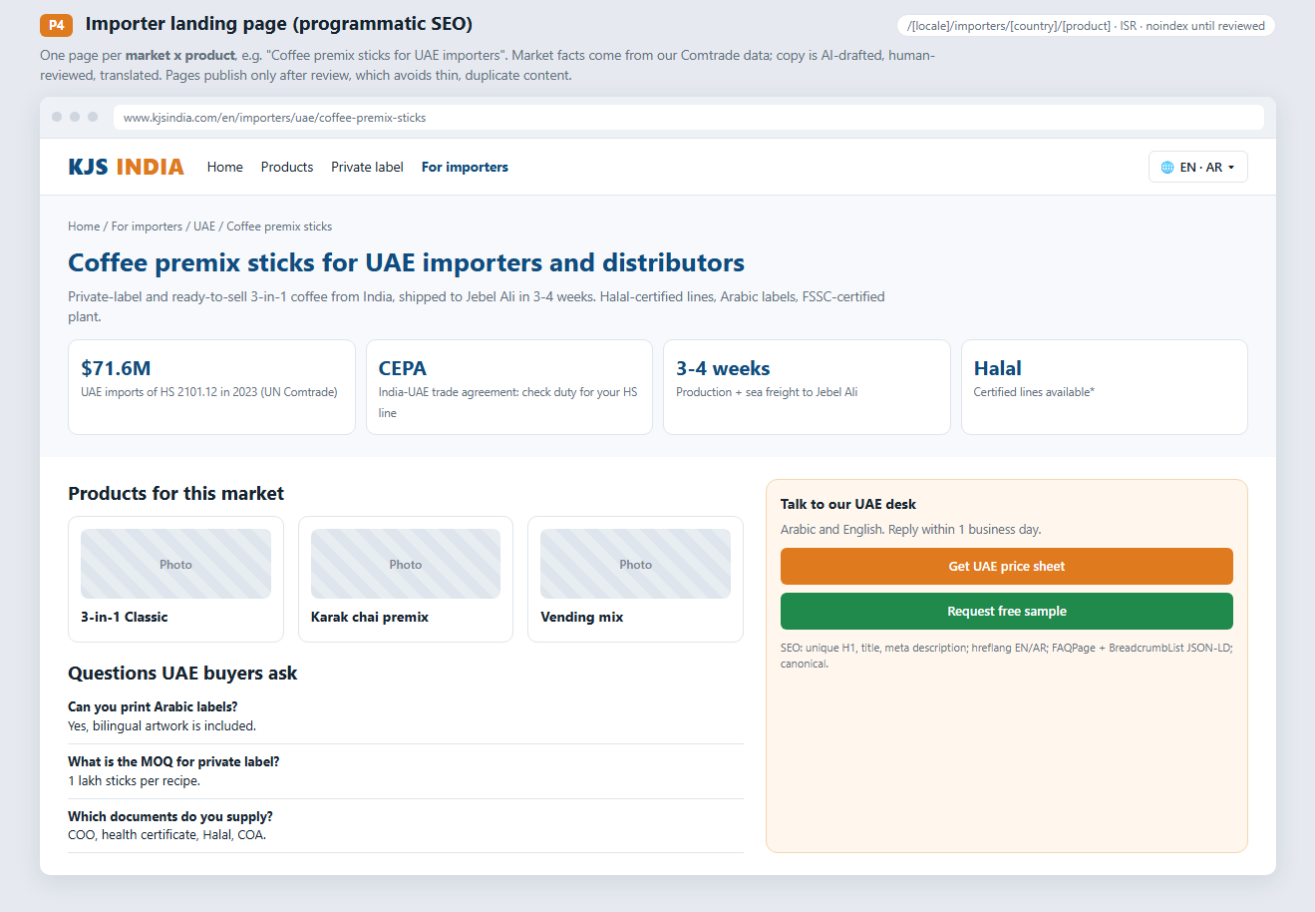
P1 Home: one-stop for importers - buyer finder, best sellers, private-label band, categories, markets, Koko.



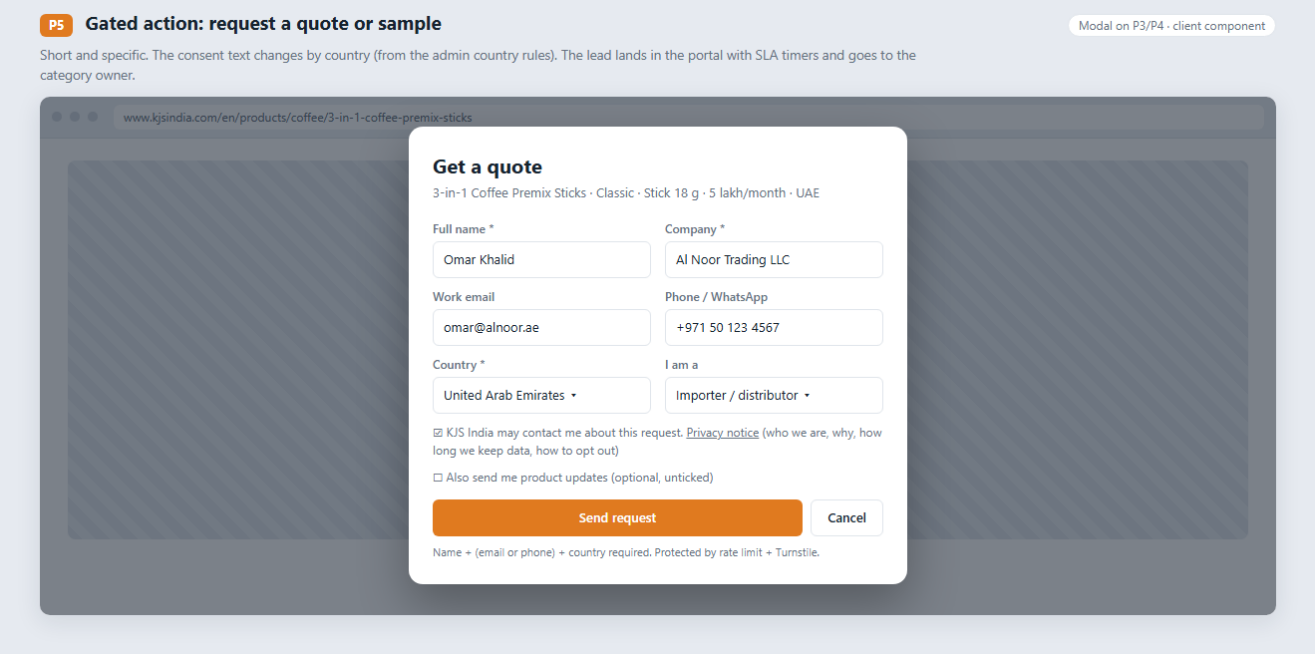
P2 Category page - open, indexable, unique intro, filters.



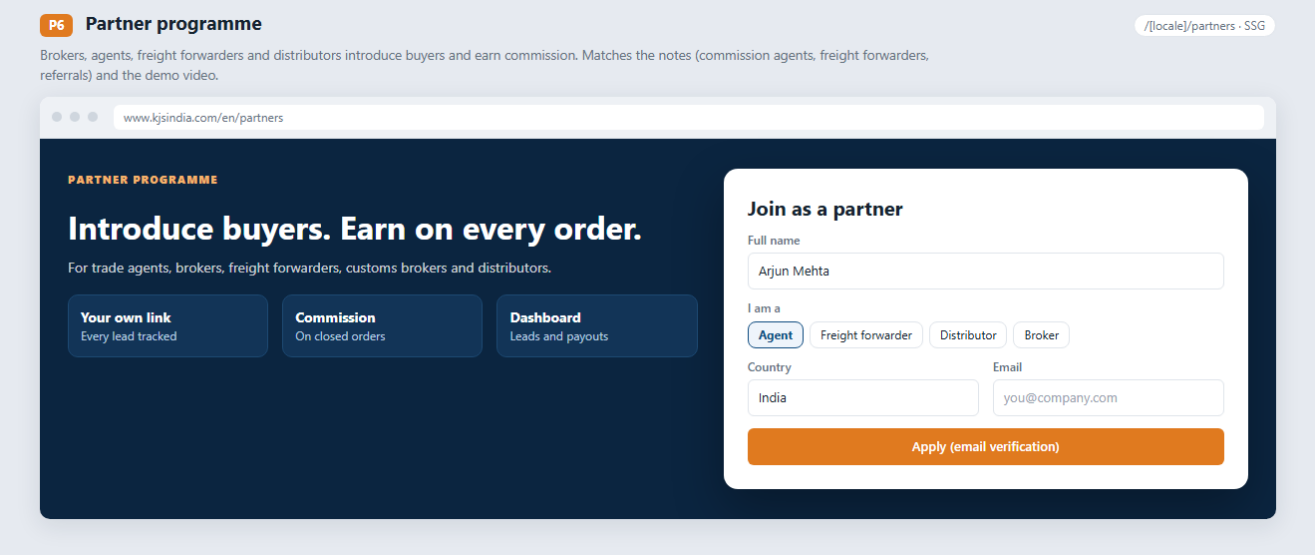
P3 Product page - public specs, gated actions, private-label configurator with indicative price band.



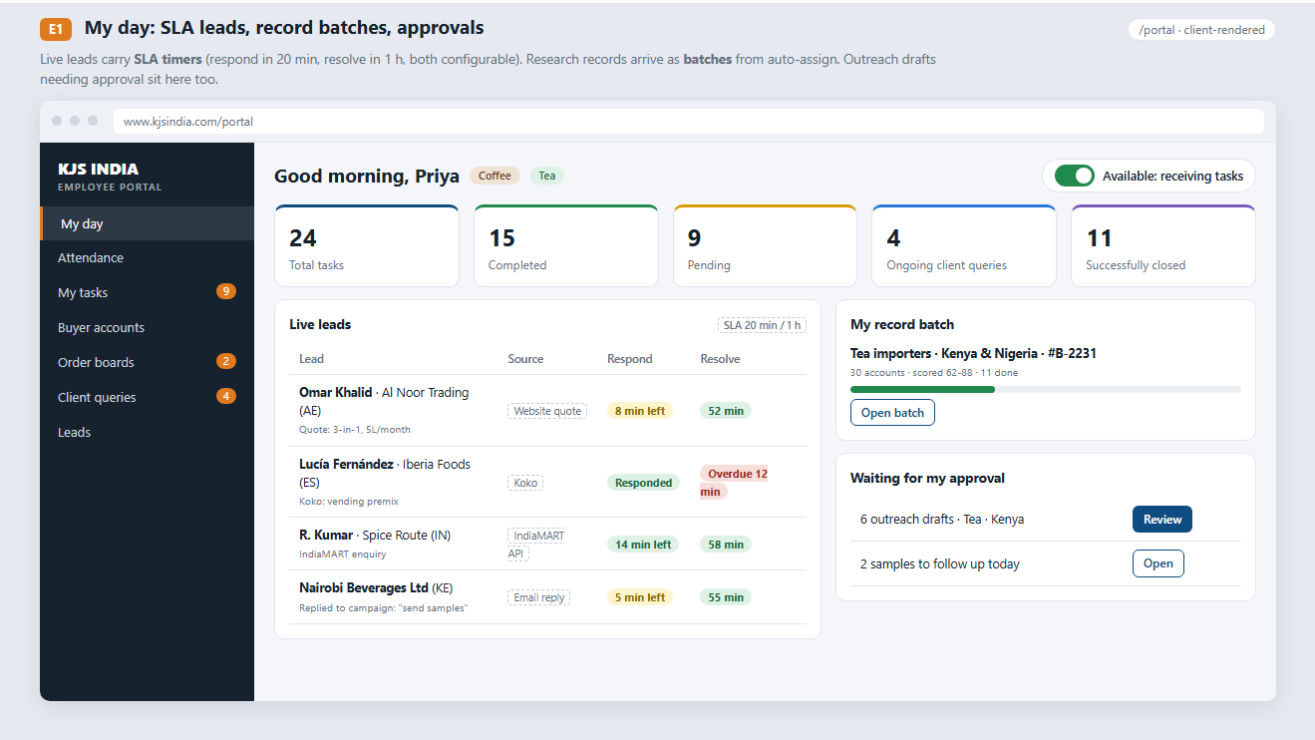
P4 Importer landing page - market x product, Comtrade facts, FAQs, hreflang, noindex until reviewed.



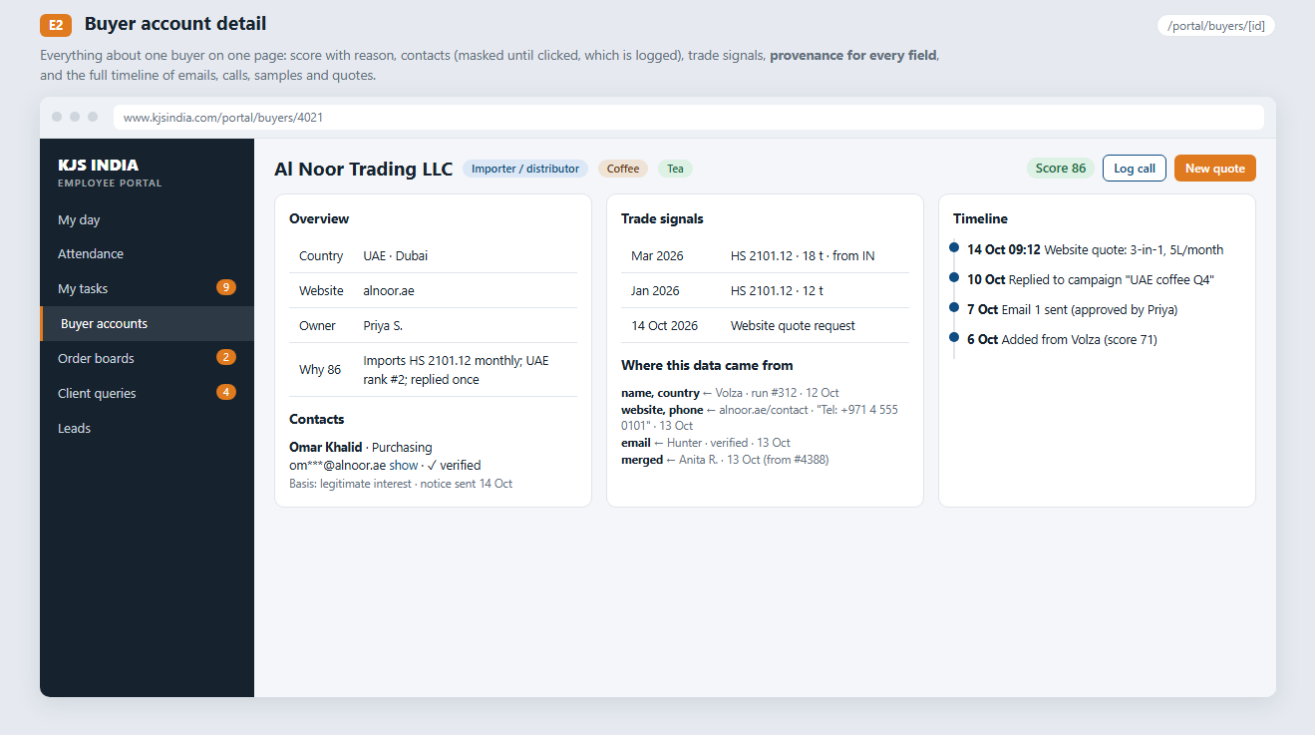
P5 Gated action modal - consent text by country, rate limit + Turnstile.



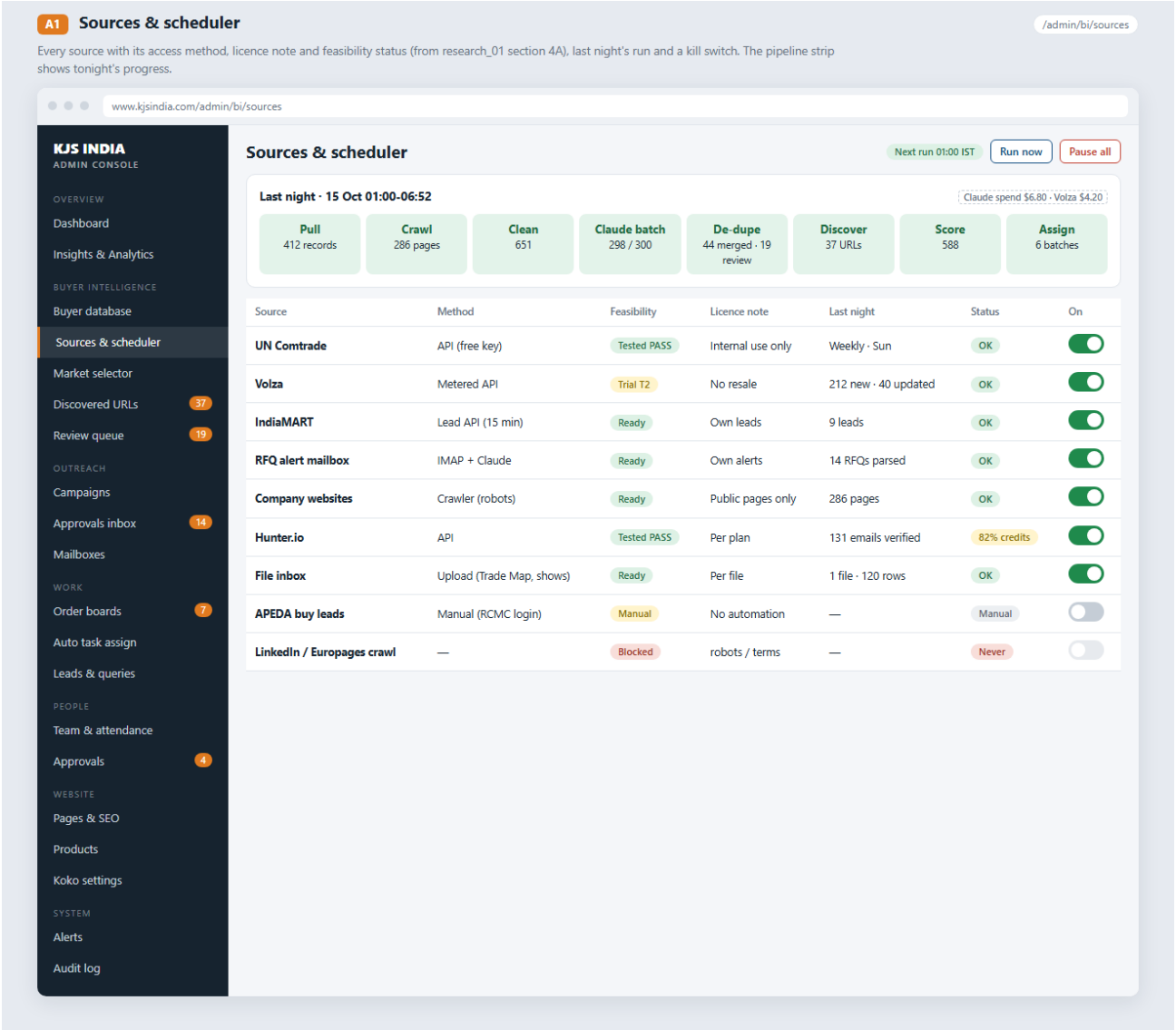
P6 Partner programme - agents, forwarders, distributors earn commission.



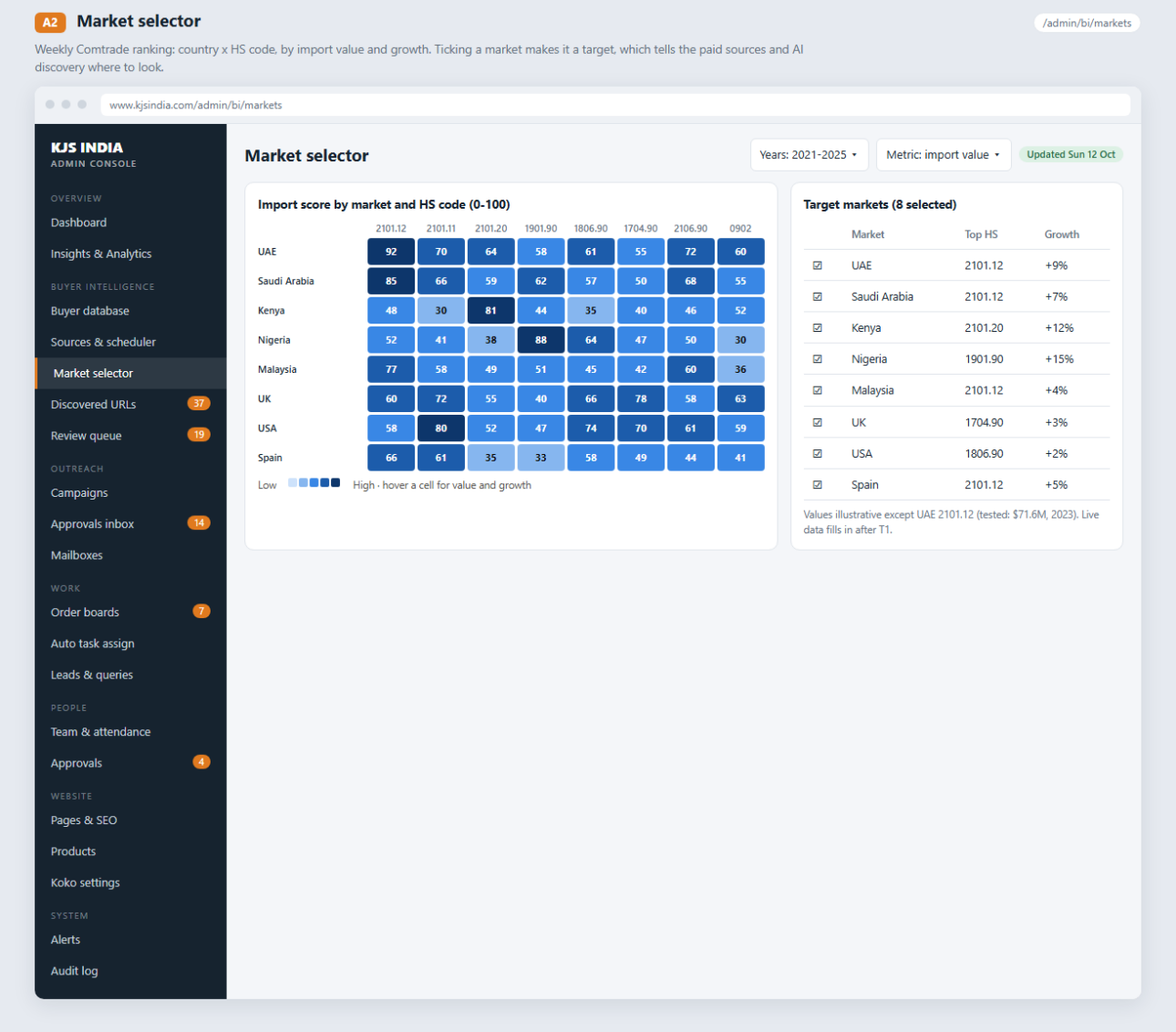
E1 My day - SLA timers on live leads, record batch, approvals waiting.



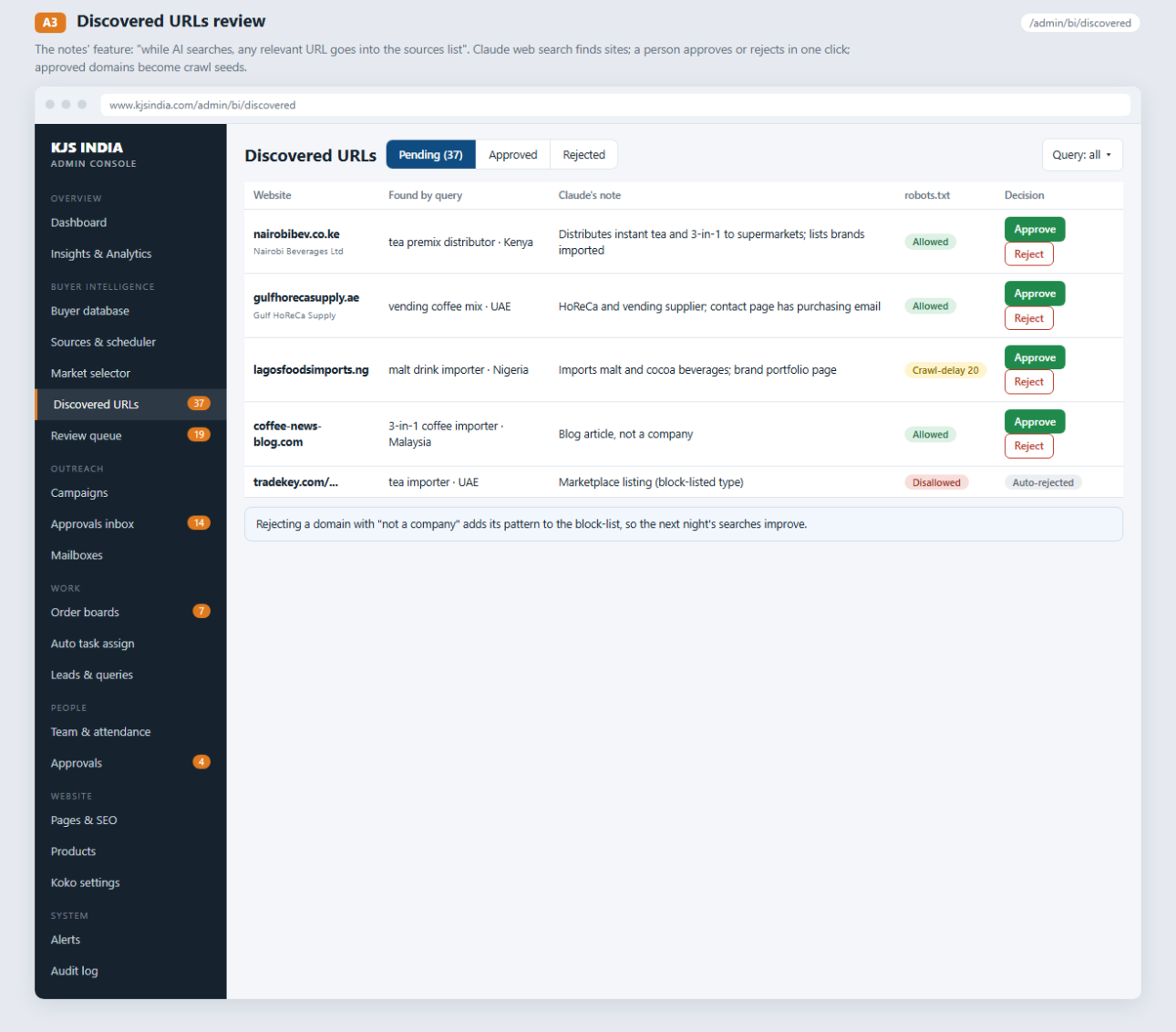
E2 Buyer account - score reason, masked contacts, trade signals, provenance, timeline.



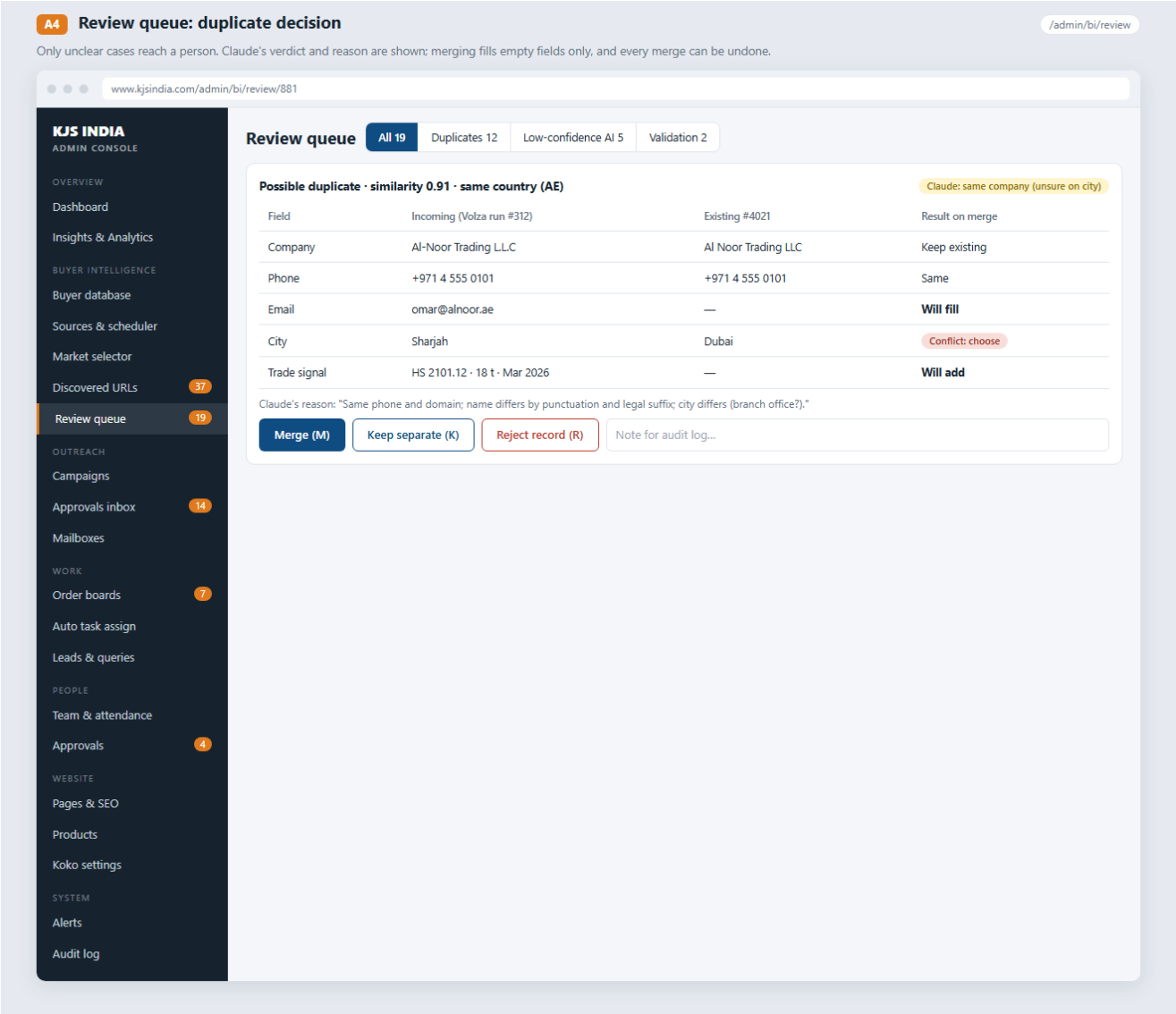
A1 Sources & scheduler - feasibility status, licence notes, nightly pipeline strip, kill switch.



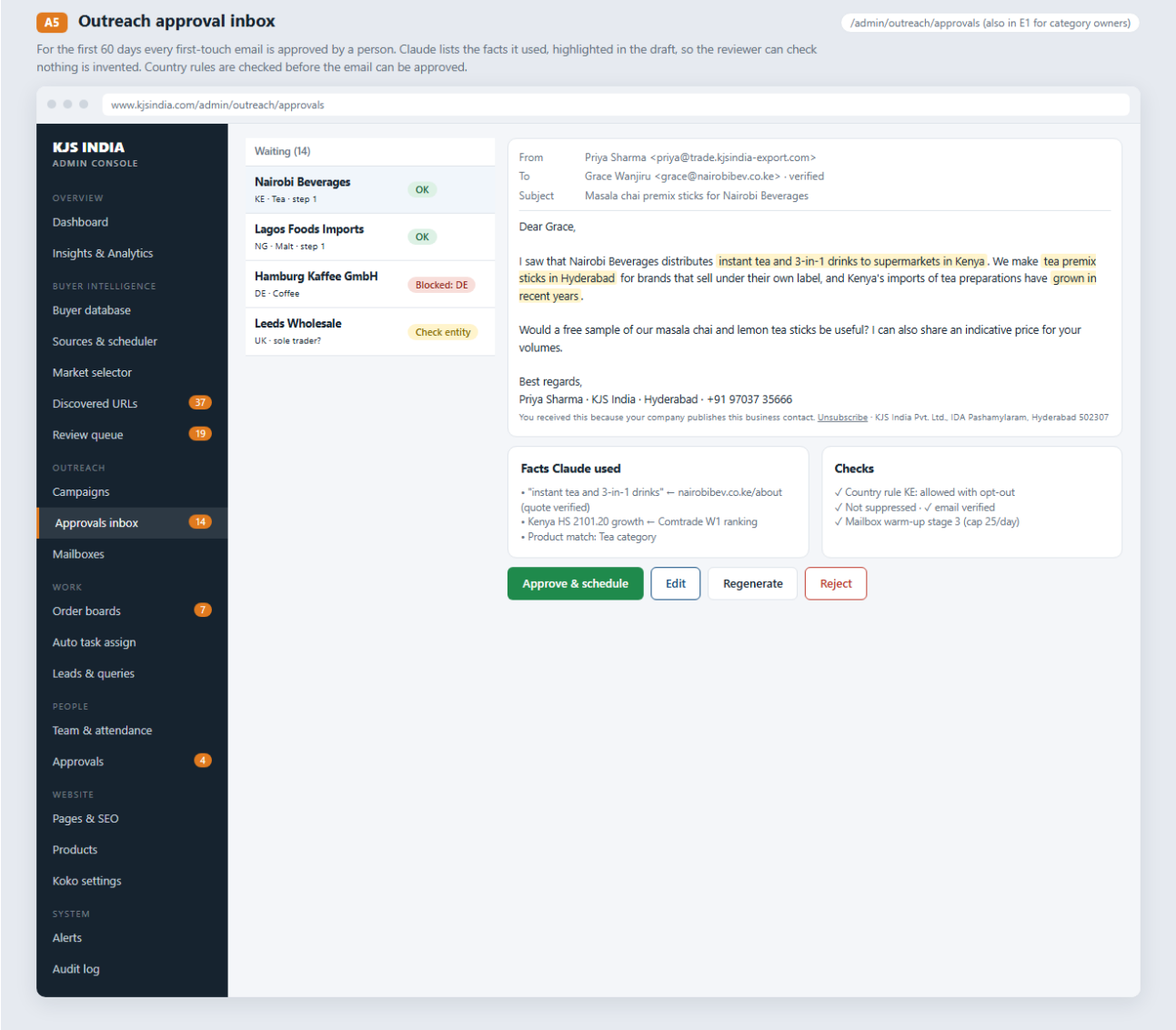
A2 Market selector - HS x country heatmap from Comtrade, target market list.



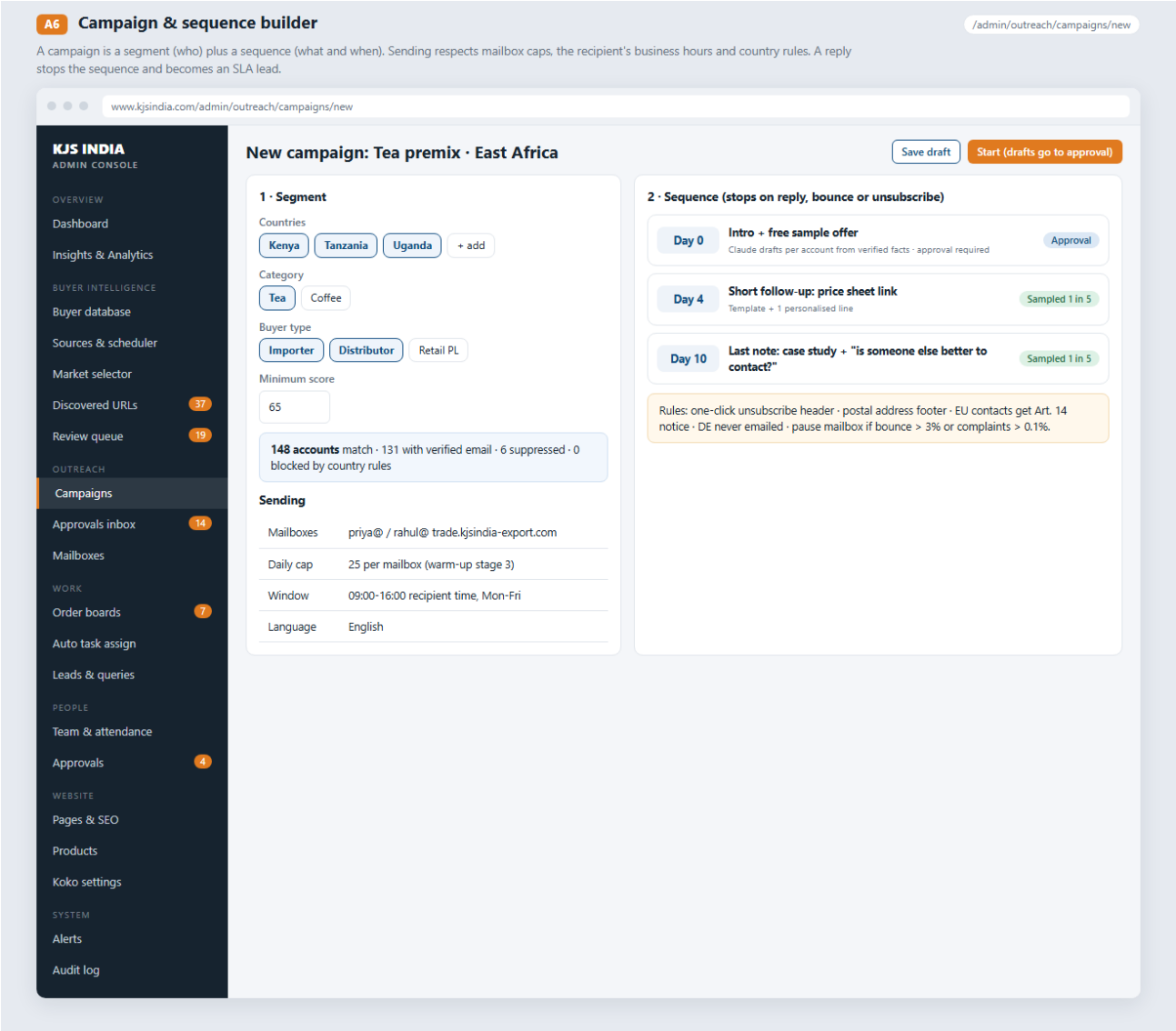
A3 Discovered URLs - AI-found sites approved or rejected in one click.



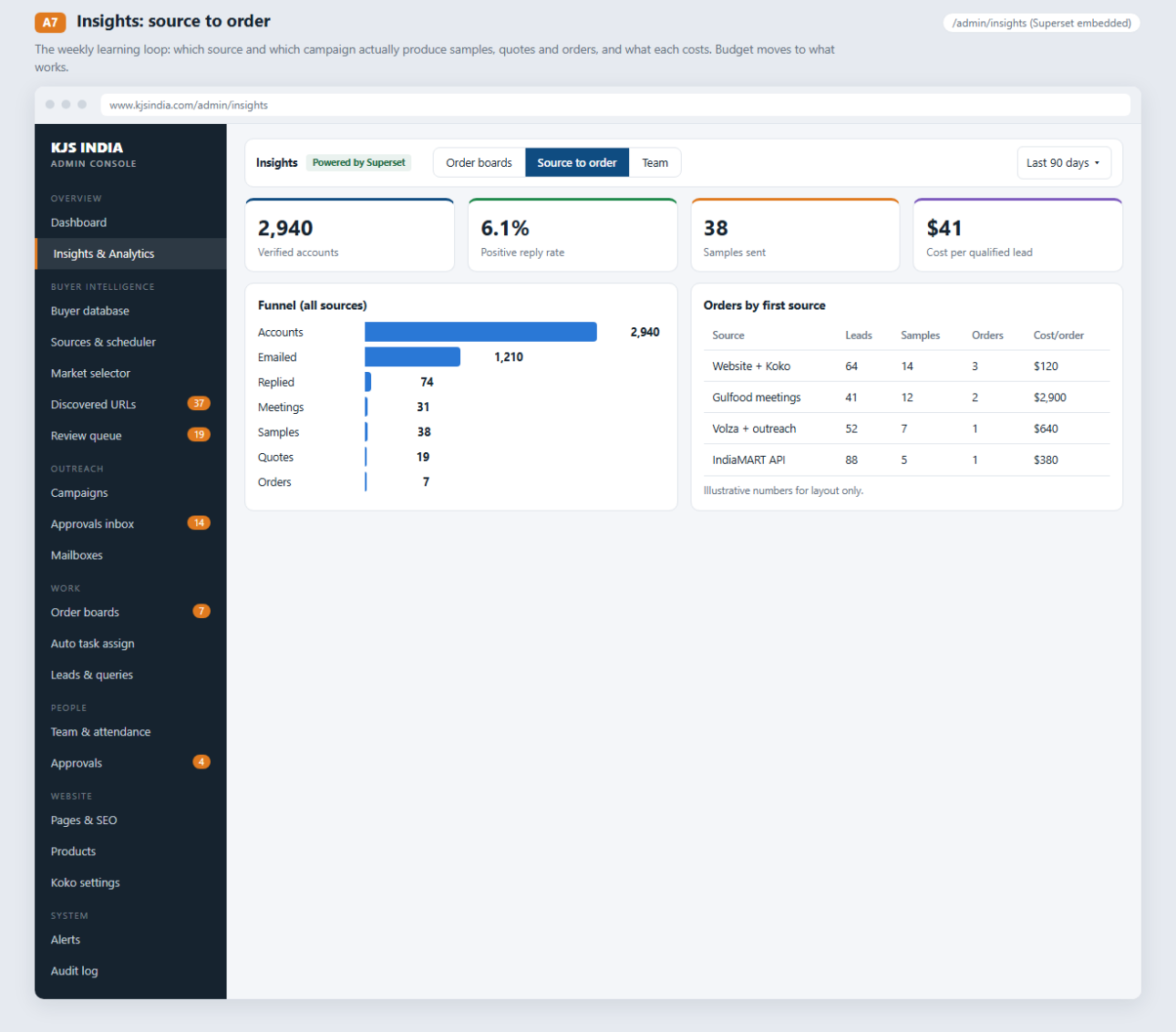
A4 Review queue - duplicate decision with Claude verdict, fill-empty merge.



A5 Outreach approval inbox - highlighted facts, country checks, approve/edit/reject.



A6 Campaign builder - segment + 3-step sequence, caps, sending window.



A7 Insights - funnel and orders by first source (Superset).

8 Backend: module layout and conventions

Layer	File	Rule
Router	modules/x/router.py	HTTP only: parse, call service, return schema. Every route declares a role dependency.
Schemas	modules/x/schemas.py	Pydantic in/out models; these generate the TypeScript client.
Models	modules/x/models.py	SQLAlchemy tables; one Alembic migration per change; never edit an applied migration.
Service	modules/x/service.py	Business rules and transactions; writes audit_log in the same transaction.
Jobs	jobs/tasks/*.py	Idempotent, checkpointed, budgeted; registered in jobs/registry.py.
Tests	modules/x/tests/	Allowed + forbidden role case per endpoint; unit tests for rules.

- Errors return a stable JSON shape {code, message, details, request_id}.
- Pagination: cursor-based for large lists (buyers, records); page/size for small ones.
- Times stored UTC; API returns ISO 8601 with offset; UI shows IST (or the viewer's zone for customers).
- Secrets only from environment; .env never committed; API keys for Anthropic, Volza, Hunter, IndiaMART, WhatsApp kept in the server secret store.

9 Data model

Consolidated list (Stage 1 from plan v1.1, Stage 2 from research). Every table has id, created_at, updated_at.

Module	Tables
Identity	users(role, status, is_available, shift_id, languages[]) · sessions · permissions · availability_log
Website	categories · products(category_id, slug, attributes JSONB, is_best_seller, rank, private_label) · product_i18n(lang, name, description, status) · price_bands(product_id, pack, qty_band, min_usd, max_usd per 1,000) · offers · pages(slug, kind, seo JSONB) · landing_pages(country, product_id, status[draft|review|published], copy JSONB, facts JSONB, noindex)
Leads & Koko	visitor_profiles(cookie_id, consent_at, country) · leads(source, company, contact, product_id, qty, country, status, owner_id, sla_respond_at, sla_resolve_at, responded_at, resolved_at) · kb_documents · kb_chunks(embedding vector(384)) · chat_sessions · chat_messages
Attendance	shifts · attendance_sessions · attendance_days · regularization_requests · holidays
Tasks & assign	tasks(source, batch_id, assigned_to, due_at, status, outcome) · task_notes · auto_assign_settings · employee_categories · employee_assign_prefs · assignment_batches · batch_records
Order boards	order_boards(code "ABC117-x") · board_steps · board_step_members · step_task_templates · step_template_items · board_step_tasks · board_documents · delete_requests
Alerts & audit	alert_rules · alert_log (unique per step/threshold/channel/recipient) · audit_log (append-only trigger) · approvals
BI sources	sources(kind, feasibility_status, licence_note, config) · job_runs · source_runs · raw_records(sha256 unique) · files_inbox · crawl_frontier · robots_cache · pages · discovered_urls
BI market	hs_codes · market_stats(reporter, hs6, period, value_usd, qty) · market_scores · target_markets
BI buyers	companies(name_key, domain, country, buyer_type, categories[], fit_score, intent_score, owner_id) · contacts(email_status, consent_basis, notice_sent_at) · trade_signals · field_provenance · review_items · merge_log
AI	ai_batches · ai_items · prompt_versions · eval_runs
Outreach	mailboxes(warmup_stage, daily_cap) · sequences · campaigns(segment JSONB) · enrollments · messages · suppression · country_rules
Conversion	samples · quotes(version, items JSONB, incoterm, valid_until, pdf_path) · partners · partner_referrals · commissions

10 API endpoints

Group	Endpoints (prefix /api)
Public (service token from Next.js)	GET public/home · public/categories · public/categories/{slug} · public/products/{slug} · public/landing/{country}/{product} · public/sitemap · public/offers
Public (browser)	POST public/gate (name, email|phone, country, consent) -> visitor cookie · POST public/quote-requests · POST public/sample-requests · POST public/configurator/price · POST koko/chat (stream) · POST partners/apply
Auth	POST auth/login · POST auth/logout · GET auth/me · POST auth/signup (staff, pending) · POST auth/password-reset
Portal	GET me/day · PATCH me/availability · attendance/check-in|check-out|week|regularize · tasks CRUD + notes · leads/{id} respond|resolve · buyers search + {id} + reveal-contact · batches/{id} records · order-boards (member view)
Admin core	dashboard · team · approvals · auto-assign settings/preview/run · order-boards CRUD/steps/members/templates · alerts rules · audit (filter, export) · website: offers, products, landing pages (publish), koko settings
Admin BI	bi/sources CRUD + run + pause · bi/runs · bi/markets (scores, targets) · bi/discovered approve|reject · bi/review/{id} merge|keep|reject · bi/buyers export (permission)
Admin outreach	outreach/mailboxes · sequences · campaigns (create, preview segment, start, pause) · approvals list/approve/edit/reject · suppression · country-rules

Group	Endpoints (prefix /api)
Webhooks	POST hooks/indiamart (push) · POST hooks/whatsapp (status) · GET unsubscribe/{token} + POST (one-click)
Internal	POST internal/revalidate-request (to Next.js) · GET health · GET health/ready

11 Business logic (algorithms)

11.1 SLA timers

```
# 11.1 Lead SLA (website, Koko, IndiaMART, email reply)
on lead created:
    owner = assign_by_category(lead.product.category)   # available owner with fewest open leads
    lead.sla_respond_at = now + settings.respond_minutes (default 20)
    lead.sla_resolve_at = now + settings.resolve_minutes (default 60)
every minute (job sla_watch):
    for lead overdue on respond: notify owner (in-app); at +10 min notify manager
    for lead overdue on resolve: mark overdue; include in dashboard and daily digest
responded_at = first outbound reply/call logged; resolved_at = outcome set
```

11.2 Auto-assign batches

```
# 11.2 Record batches (auto-assign)
every 15 min (Mon-Sat, working hours) and on events:
    free = users active & available & no open batch & batches_today < max
    for emp in free (longest waiting first):
        rows = SELECT id FROM companies WHERE category IN emp.categories
               AND owner_id IS NULL AND status='qualified'
               ORDER BY fit_score DESC LIMIT emp.batch_size FOR UPDATE SKIP LOCKED
        create batch + task; set owner_id; audit
```

11.3 Order boards

```
# 11.3 Order board step timing
W = working_days(start, deadline)             # Mon-Sat minus holidays
step_days[i] = W * pct[i]/100  (or manual days)
planned_end[i] = add_working_days(planned_start[i], step_days[i])
elapsed_pct = worked_time_since_start / step_time * 100
alerts at 50 / 85 / 100 % -> mail + WhatsApp, once each (unique alert_log)
```

11.4 Buyer pipeline

```
# 11.4 Buyer pipeline per record
raw -> normalise(country, phone, email, name_key, domain)
    -> extract (Claude batch, JSON + quotes) -> keep field only if quote in source
    -> dedupe: exact(domain|email|source_id) -> fuzzy(pg_trgm>=0.85, same country)
               -> Claude judge -> auto-merge if 'same' + phone/city match else review
    -> score = 0.6*fit + 0.4*intent  (capped by penalties) + one-line reason
    -> status 'qualified' when score >= threshold and a verified contact exists
```

11.5 Outreach

```
# 11.5 Outreach enrollment state machine
enrolled -> drafted (Claude) -> approved (human) -> scheduled (recipient window, mailbox cap)
         -> sent -> [replied | bounced | unsubscribed | completed]
before 'approved': country_rule allows? not suppressed? email verified? else blocked
on reply: Claude classifies -> interested: lead + SLA ; unsubscribe: suppression ;
          later: snooze 30 days ; out-of-office: retry after return date
mailbox guard: bounce > 3% or complaints > 0.1% in 7 days -> pause mailbox + alert
```

11.6 Gate

```
# 11.6 Gate and consent
first gated action -> modal (name, email|phone, country, consent text by country)
POST public/gate -> visitor_profile + signed cookie (1 year) + lead
later gated actions -> no modal; lead updated; same owner
withdraw/erase request -> suppression + erase personal fields + audit (DPDP/GDPR)
```

12 Jobs and nightly scheduler

Time (IST)	Job	Purpose
Every 1 min	sla_watch	SLA breaches and escalations
Every 15 min	indiamart_pull, rfq_mailbox, reply_classify, batch_assign, deadline_alerts	Daytime flow
01:00	pull_sources	Volza (budgeted), file inbox, catch-up of IndiaMART and RFQ mail
01:30	crawl	RSS/sitemaps, then allowed pages; robots + rate limits
02:00	pipeline	Clean, validate, pre-filter
02:30	ai_batch_submit	Claude batches: extract, classify, dedupe judge, drafts
03:00	backup	pg_dump + files, encrypted, off-site
03:15-06:45	ai_batch_collect	Every 10 min; verify quotes; write provenance
04:30	dedupe_merge	Exact, fuzzy, judged merges; review queue
05:00	ai_discovery	Claude web search for target markets
05:30 / 06:00	enrich / score	Hunter verification; fit + intent
06:30	outreach_queue	Schedule approved emails in recipient hours
07:00	assign_and_digest	Batches to owners; morning digest with spend
Sun 01:00 / Mon 07:30	market_rank / attribution	Comtrade refresh; source-to-order report
Daily 04:00	sitemap_refresh	Rebuild sitemap; revalidate changed pages

Rules: job_runs UNIQUE(job, run_date); checkpoints; 3 retries with backoff then dead-letter + alert; per-job budgets; kill switch per source and for outreach.

13 Claude (Anthropic API) design

Task	Surface	Model	Effort	Output
Koko chat (RAG, multilingual, lead capture)	Messages, streaming, prompt caching	claude-opus-5-5	low	Text + tool calls (create_lead)
Extraction from pages / PDFs / RFQ emails	Message Batches	claude-opus-5-5	low	JSON schema + source quotes
Classification, dedupe judge	Message Batches	claude-opus-5-5	low / medium	JSON enums + reason
Outreach drafts	Message Batches	claude-opus-5-5	medium	Subject, body, facts used
Landing page copy + translation drafts	Message Batches	claude-opus-5-5	medium	JSON per locale; human review
AI URL discovery	Messages + web_search_20260209 (blocked domains)	claude-opus-5-5	medium	URLs to review queue
Reply classification, card photos	Messages + server-side fallback	claude-opus-5-5	low	JSON

- One service module (app/ai/client.py) for all calls: model, effort, max_tokens, prompt version, token and $ logging per call.
- Batches: requests keyed by custom_id; results read in any order; refused items retried next day as single calls with the server-side fallback beta (fallbacks="default").
- Structured outputs (output_config.format json_schema) for every machine-read answer; parse with json.loads, never string matching.

- Untrusted page/email text is passed as data with an instruction to ignore instructions inside it.
- Costs: ~$7/night batch work + Koko and daytime calls, ~$270-330/month on claude-opus-5-5; monthly limit in the Anthropic Console. A cheaper model for bulk extraction (claude-haiku-5-5) is a management decision after the 50-item golden-set comparison.
- Code samples for batch submit/collect, web search and fallback: research_01_implementation_plan.pdf section 6 (unchanged).

14 Source connectors and feasibility

Connector	Status (research_01 section 4A)	Build in
UN Comtrade API	Use now - tested PASS	BI Sprint 2
IndiaMART Lead API (pull + push)	Use now - needs paid seller key (T5)	BI Sprint 3
RFQ alert mailbox (Alibaba, Tridge, Europages, FIEO)	Email parse only (T6)	BI Sprint 3
Hunter.io API	Use now - tested PASS (T9)	BI Sprint 4
Volza API	Trial first (T2); build connector only after pass	BI Sprint 4
File inbox (Trade Map exports, show lists, Kompass sample)	Use now	BI Sprint 1
Google Places (HoReCa, place_id only)	Pilot (T10)	Phase 3
APEDA buy leads	Manual (RCMC login) - no connector	-
LinkedIn, Europages crawl, Alibaba/IndiaMART scraping	Blocked - never built	-

15 Security, privacy, compliance

Area	Control
Auth	Argon2 passwords; httpOnly Secure SameSite=Lax session cookie; CSRF token header; login rate limit; staff approval before access
Authorisation	Role + permission per route (admin, manager, employee, partner); contact reveal logged; exports need permission
Public forms	Rate limit per IP, honeypot, Cloudflare Turnstile, server-side validation
Data protection	Consent/notice text by country (country_rules); source logged per contact; erasure workflow; retention: chat 90 days, rejected leads 12 months
Outreach law	DE blocked; UK companies only; EU Art.14 notice; US address + opt-out; WhatsApp only with opt-in
Audit	append-only audit_log for every write that matters (trigger blocks update/delete)
Secrets & network	Env/secret store; no ports open (Cloudflare Tunnel); SSH via Tailscale; Cloudflare Access on staging and Superset
Backups	Nightly encrypted off-site (restic), 7 daily / 4 weekly / 6 monthly; monthly restore drill

16 DevOps, environments, monitoring

- Environments: local (docker compose), staging (staging.kjsindia.com behind Cloudflare Access), production. Same images, different .env.
- CI (GitHub Actions): ruff + mypy + pytest; eslint + tsc + vitest; next build; OpenAPI client regenerated and diffed; Playwright E2E on staging.
- Deploy: tag -> build images -> push -> deploy.sh on server (migrate, then rolling restart); keep last 3 image tags for rollback.
- Monitoring: Sentry (web + api), UptimeRobot on /api/health and home, morning digest (jobs, spend, SLA), Lighthouse CI on public pages.

17 Testing strategy

Level	What	Gate
Unit	Normalisers, de-dupe, scoring, SLA, order-board maths, country rules, batch assign	CI must pass
API	Every endpoint: allowed + forbidden role, validation errors	CI must pass
Contract	Connectors with recorded responses (no live calls in CI)	CI must pass
AI evals	Golden sets (50 items) per Claude task; accuracy stored per prompt version	No regression
E2E	Browse -> gate -> quote -> SLA lead; staff attendance + tasks; board -> alert; upload -> review -> buyer; draft -> approve -> send (sandbox) -> reply -> lead	Before each deploy
SEO	Lighthouse >= 90, valid JSON-LD, hreflang pairs, sitemap URLs return 200	Before public release

18 Delivery plan with dates

Stage 1 - Platform (go-live Fri 11 Dec 2026)

Week	Dates	Build	Done when
0	15-16 Oct	Docs frozen (this plan, research_01, ui v3); repo; CLAUDE.md; accounts (Anthropic, Cloudflare, WhatsApp verification started)	Repo + CI green
1	19-23 Oct	FastAPI core, auth, roles, audit; Next.js app with next-intl, layouts, design tokens, sidebar config; staging via Tunnel	Login works on staging
2	26-30 Oct	Public P1-P3 + P5 gate, offers ticker, sitemap/robots/metadata/JSON-LD; admin content CRUD + revalidation	Lighthouse >= 90 on staging
3	2-6 Nov	Portal: attendance, availability, tasks + notes + own tasks, dashboard stats; leads with SLA (E1)	Staff use it for a full day
4	9-13 Nov	Auto-assign batches; Koko v1 (RAG, widget, lead capture); P6 partners	Koko creates SLA leads
5	16-20 Nov	Order boards core + step templates + files	A board runs 5 steps
6	23-27 Nov	Approvals, delete requests, alerts (mail + WhatsApp)	Alerts fire once at 50/85/100%
7	30 Nov-4 Dec	Superset embed; hardening; backups drill; E2E; SEO audit	All gates green
8	7-11 Dec	UAT, fixes, production go-live	Live on www.kjsindia.com

Stage 2 - Buyer Intelligence & Outreach (2027, gated)

Sprint	Dates	Build (screens)	Gate / done when
S1	4-15 Jan	Sources registry, job_runs, file inbox, run monitor (A1)	File becomes raw_records
S2	18-29 Jan	Comtrade connector, market scores (A2), normalisers, crawler core	Top-15 markets ranked

Sprint	Dates	Build (screens)	Gate / done when
S3	1-12 Feb	Claude extraction batches, review queue (A4), IndiaMART API, RFQ mailbox; start mailbox warm-up	Golden-set accuracy measured
S4	15-26 Feb	Volza (if T2 passed), Hunter, de-dupe, scoring, discovery (A3), buyer UI (E2)	Gate 1: 1,500 verified accounts
S5-S6	1-26 Mar	Mailer, sequences (A6), approvals (A5), reply -> SLA lead, unsubscribe, country rules; Gulfood follow-ups	Approved emails flow
S7-S8	29 Mar-23 Apr	P4 landing pages EN/AR/ES/FR, configurator price bands, attribution (A7), DPDP erasure	Gate 2: replies above baseline
S9-S12	26 Apr-18 Jun	Samples, quotes, trade-show mode (before PLMA 25-26 May), partner portal, hardening	Gate 3: first attributed orders

19 Claude Code task prompts

Use one prompt per task, in plan mode first. Template:

```
Task: <one sentence>  (screen <ID> in docs/research/ui_designs_v3.html)
Context: docs/PRD.md section <x>; implementation_plan_v2 sections <y>; data model tables <...>
Acceptance criteria:
  1. <API endpoint + role rule>
  2. <UI behaviour matching the screen>
  3. <audit / SLA / consent rule>
  4. Tests: allowed + forbidden role; unit tests for <rule>; E2E path <...>
Constraints: follow CLAUDE.md; no new dependency without asking; Next.js public pages
must stay server-rendered; regenerate the API client after schema changes.
First give me a plan (files, migrations, endpoints, components, tests). No code yet.
```

Order	Example first tasks
1	Monorepo skeleton: Next.js (App Router, TS, Tailwind, shadcn, next-intl) + FastAPI + Postgres in docker compose; /api/health; CI
2	Auth: staff sign-up -> pending -> admin approve; login/logout; /auth/me; Next.js middleware guard for /portal and /admin
3	Public home P1 as server component with ISR tag "home" + generateMetadata + Organization JSON-LD
4	Gate P5: POST /public/gate with country consent text; visitor cookie; lead with SLA timers
5	Shared sidebar config by role; portal E1 My day reading /me/day

20 Open items

Item	Owner	By
Approve Next.js switch and the +3 day Week-1 impact	Management	16 Oct 2026
Brand assets (logo, colours, photos) and product list with price bands for the configurator	KJS	23 Oct 2026
Public languages confirmed (EN, AR, ES, FR)	Management	23 Oct 2026
Anthropic production key + monthly limit; WhatsApp Business verification	Management / IT	23 Oct 2026
Category gate decision (actions only vs form before categories)	Management	23 Oct 2026
Paid source trials T2 (Volza), T5 (IndiaMART), T8 (Kompass)	Management	Dec 2026
Sending domain + mailboxes for outreach	IT	Jan 2027